

EXTERN × WAYFAIR

Wayfair Rugs Market Intelligence

AI Agent Demo

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Category analysed: Shag Rug



Moodboard Generator output

Executive Summary

Four connected agents feeding one Market Intelligence Dashboard

OBJECTIVE

Give Wayfair's Rugs team an automated research pipeline for Shag Rugs — one that finds trends, benchmarks competitors, and produces decision-ready recommendations without an analyst spending a day clicking through listings.

MY SOLUTION

Built 4 AI agents in n8n. Stages 1–2 validate input and collect data with no LLM calls; AI reasoning enters at Stage 3 with role-specific system prompts. All outputs converge in a six-tab dashboard.

CORE OUTPUTS

[Moodboard Generator](#)

[Trend Discovery Agent → Trend Report](#)

[Competitor Monitoring Agent → Report](#)

[AI Insights & Content Agent → Report](#)

[Integrated Dashboard](#)

KEY INSIGHTS

- 89% of top-performing shag rugs use beige, gray or cream — neutral high-pile shag is now modern minimalism, not retro revival
- Mid-range \$100–\$500 captures 56% of the market; 8'x10' and 5'x7'6" polyester dominate
- \$400+ premium is virtually uncontested — one product above \$300 at each of the three retailers
- Ultra-plush faux fur under \$50 is rising in bedrooms and nurseries but is only 11% of assortment

FUTURE IMPROVEMENTS

Connect internal SKU and margin data so recommendations can be sized in revenue rather than price bands, and replace the mock Instagram and Pinterest routes with licensed social APIs.

Agent 1 — Moodboard Generator

From design prompt → AI-curated visual mood

PROJECT 1

OBJECTIVE

Turn a short style phrase into a polished moodboard, so the design team can see a trend direction in seconds instead of briefing a shoot.

INPUT PROMPT EXAMPLE

“bohemian rugs, neutral tones”

KEEP IN MIND / INPUT NOTES

- Keep input under 10 words and specific — name the material and style, not the mood
- A Clean Prompt node strips line breaks, Markdown and double quotes before the image call; unescaped characters break the HTTP request
- HuggingFace returns the binary with a broken filename, so a code node rebuilds it as moodboard.png with the correct MIME type — without this the download will not open
- Free-tier image generation is rate-limited per day

IMPROVEMENTS

Pin the seed value so outputs are reproducible — Wayfair needs consistent on-brand visuals, not random experiments. A higher-tier API would allow product-aligned rendering.



Output example — neutral-tone bohemian moodboard

[View JSON Workflow](#)

Agent 2 — Trend Discovery

From category & market signals → decision-ready trend report

PROJECT 2

OBJECTIVE

Automate shag-rug trend research end to end — real product data, social signals and industry commentary, turned into a structured report the category team can act on.

INPUT PROMPT EXAMPLE

Category: Shag Rug | focus: washable

HOW IT WORKS

- Stage 1 validates the category and exits early with a clear error if none is detected
- Stage 2A fetches Amazon products; Stage 2B runs in parallel for Instagram, Pinterest, blogs and market forecasts
- Stages 3–6 classify products, extract attributes, identify micro-segments and write each report section

IMPROVEMENTS

Extend to further rug categories and sources, and add historical storage so trends read as change over time rather than a single snapshot.

89%

of top performers in
beige / gray / cream

56%

of market sits in the
\$100–\$500 band

35

signals pulled across
4 source types

KEEP IN MIND / INPUT NOTES

- Instagram and Pinterest run on mock API routes — live scraping breaches Meta and Pinterest terms. The live Architectural Digest RSS feed is the real-time driver.
- RSS sources were switched from lifestyle blogs to publication-grade design sources (Architectural Digest, Dezeen) for design-heavy insight
- A two-tier filter strips sub-categories to their root noun (“shag rug” → “rug”) and requires a design-context word before admitting an article
- Excerpts truncate at 600 characters — enough depth on texture and colour without inflating token cost

[View JSON Workflow](#)

[View Trend Report](#)

Agent 3 — Competitor Monitoring

From competitor listings → pricing and whitespace intelligence

PROJECT 3

OBJECTIVE

Benchmark Wayfair against Amazon and Walmart on price, assortment, ratings and features, and surface the gaps worth entering.

INPUT PROMPT EXAMPLE

Category: Shag Rug | Amazon + Walmart collection & product URLs

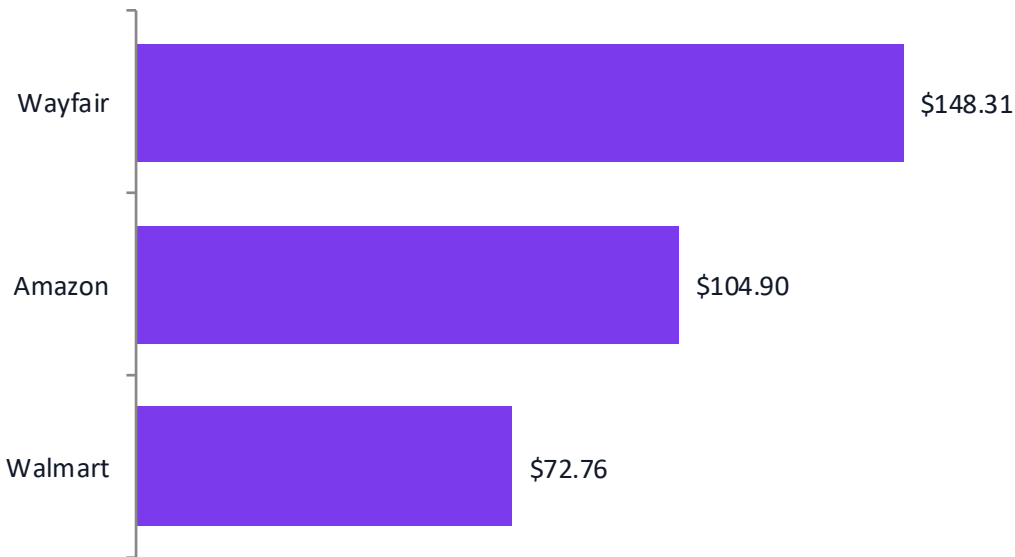
WHAT IT FOUND

- Wayfair sits at the TOP of the market: \$148.31 average vs Amazon \$104.90 and Walmart \$72.76
- 70% of Wayfair's assortment is \$100–\$300; Amazon and Walmart put 60% and 70% under \$100
- Ratings are effectively level — Wayfair 4.43, Amazon and Walmart 4.51 each
- Wayfair discounts least: 80% of its products carry no discount, against Amazon's 60%
- \$400+ is open territory — one product above \$300 at each retailer

KEEP IN MIND

Based on a 30-product sample (10 per retailer), not full catalogues. Prices are a point-in-time snapshot. The generated summary called Wayfair “underpriced” — its own tables disprove that, so summaries need checking against their data.

Average shag rug price by retailer



IMPROVEMENTS

Widen the sample beyond 30 SKUs, add Target and Overstock, and schedule weekly runs so price movement is tracked rather than sampled once.

[View JSON Workflow](#)

[View Competitor Report](#)

Agent 4 — AI Insights & Content

From intelligence → prioritised, rationale-backed actions

PROJECT 4

OBJECTIVE

Convert the trend and competitor reports into ranked recommendations, each carrying a rationale field so a reviewer can trace the reasoning.

INPUT PROMPT EXAMPLE

Market Analysis + Competitor Analysis → content strategy

KEEP IN MIND

- Recommendations inherit any error upstream, so the input reports must be verified first
- Impact estimates are directional — the agent has no access to Wayfair margin or inventory data
- Every recommendation carries a rationale so the logic is auditable

IMPROVEMENTS

Connect internal SKU, margin and inventory data so recommendations can be sized in revenue and checked against what Wayfair can actually source.

OUTPUT — 7 RECOMMENDATIONS, PRIORITY-RANKED

Launch a \$400–\$800 designer luxury line

HIGH

Enters the uncontested premium band

Introduce an \$89–\$129 “Value Plus” washable line

HIGH

Blunts Walmart without cannibalising premium

Expand mid-range Neutral Shag by 30%

HIGH

Prioritise 8'x10' beige and cream polyester

Develop a \$50–\$100 faux fur tier

MED

11% of assortment against rising demand

[View JSON Workflow](#)

[View Output Report](#)

Market Intelligence Dashboard

Where all four agents converge

PROJECT 5

OBJECTIVE

One place for the category team to read category health, competitive threats and next actions without opening four separate reports.

HOW IT WORKS

The Trend report (P2) and Competitor report (P3) feed six tabs. The Executive Overview surfaces the target segment, Wayfair's average price, the headline trend and competitive insights, live risk alerts, and the top three actions.

KEEP IN MIND / IMPLEMENTATION NOTES

- Sample-based (30 products) and says so on the page — deliberate, so nobody reads it as full-catalogue truth
- Snapshot dated 2026-08-11; refresh requires re-running the upstream agents
- The supplier panel returns “No suppliers found” — that field needs a data source it does not yet have

IMPROVEMENTS

Scheduled refresh, threshold alerting on competitor price moves, and multi-category support.

SIX TABS

Executive Overview

Market & Trends

Competitive Intel

Opportunity Radar

Risk & Diagnostics

Action Center

MARKET CONTEXT SURFACED

\$62.9B

global carpet & rug
market, 2025

9.59%

CAGR to 2033
(\$130.62B)

12.4%

CAGR — premium,
fastest-growing tier

[View JSON Workflow](#)

[View Dashboard](#)

Reflections & Future Improvements

What building this actually taught me

1 Prompt engineering is iterative, and specificity is the whole game

Vague input produces useless output, and models hallucinate confidently. My own competitor report described Wayfair as “underpriced” when its own tables showed Wayfair as the most expensive of the three retailers. I learned to constrain prompts to the data in front of the model, and to check every generated summary against the numbers it claims to summarise.

2 Audit the system you are handed rather than accepting it

The starter RSS sources were lifestyle blogs returning content with no design substance, so I replaced them with Architectural Digest and Dezeen. Exact-match category filtering was silently dropping relevant articles, so I built a two-tier filter on root nouns plus a design-context word. Neither problem was flagged for me — both surfaced only by reading the outputs.

3 Design for failure, not just the happy path

Stage 1 validates the category and exits with a clear error instead of passing bad input downstream. When I hit anti-scraping restrictions on Instagram and Pinterest, I kept mock routes for those and promoted the live RSS feed to the primary source, so one blocked input could not take down the workflow.

4 Separate the plumbing from the reasoning

Stages 1 and 2 are plain JavaScript and HTTP calls with no LLM tokens; AI enters only at Stage 3. That kept cost and latency down and made failures far easier to locate.

Next steps: connect internal SKU and margin data · licence real social APIs · store history so trends read as movement · scheduled refresh with threshold alerts

Thank you

Four agents, one dashboard, and a shag rug category that now reports on itself.

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[Trend Report](#)

[Competitor Report](#)

[Insights Report](#)

[Dashboard](#)